

BLAST Sequence Analysis Report — Human Insulin (INS)

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SEQUENCE ANALYZED

Gene: INS (Insulin)

Organism: Homo sapiens

Source: NCBI Accession MT335689.1

BLAST PARAMETERS

Tool: Nucleotide BLAST (blastn)

Database: nr (non-redundant)

Website: blast.ncbi.nlm.nih.gov

PYTHON ANALYSIS RESULTS

Total Length: 751 bases

A count: 154

T count: 116

G count: 226

C count: 244

GC Content: 62.58%

Unexpected bases (N): 11

Total valid nucleotides: 740

Interpretation of Python Analysis:

- High GC content (62.58%) indicates thermally stable DNA, consistent with a highly conserved and functionally important gene
- 11 unexpected bases (N) detected, suggesting

- minor sequencing ambiguities at certain positions
- These N bases are common in real sequencing data and do not affect overall gene identification

BLAST RESULTS

Top Match: Homo sapiens insulin isoform UA (INS) mRNA

% Identity: 100%

E-value: 0.0

Query Coverage: 100%

CROSS SPECIES COMPARISON

Same sequence BLASTed against mouse database:

Top Match: Mus musculus insulin II (Ins2) mRNA

% Identity: 83%

E-value: 7e-82

Query Coverage: 71%

Interpretation of Cross Species Results:

- 83% identity between human and mouse insulin confirms strong evolutionary conservation
- 17% difference reflects millions of years of separate evolution while maintaining core insulin function
- Query coverage of 71% suggests structural differences at certain regions between species
- E-value of 7e-82 still confirms genuine biological relationship despite species difference

OVERALL CONCLUSION

Sequence conclusively identified as human insulin gene (INS). High GC content (62.58%) confirms

thermal stability expected of a functionally critical metabolic gene. Cross-species comparison with mouse insulin demonstrates strong evolutionary conservation across mammalian species, consistent with the essential role of insulin in glucose metabolism.

The 11 ambiguous bases (N) detected suggest minor sequencing quality issues but do not affect overall gene identification confidence.

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